

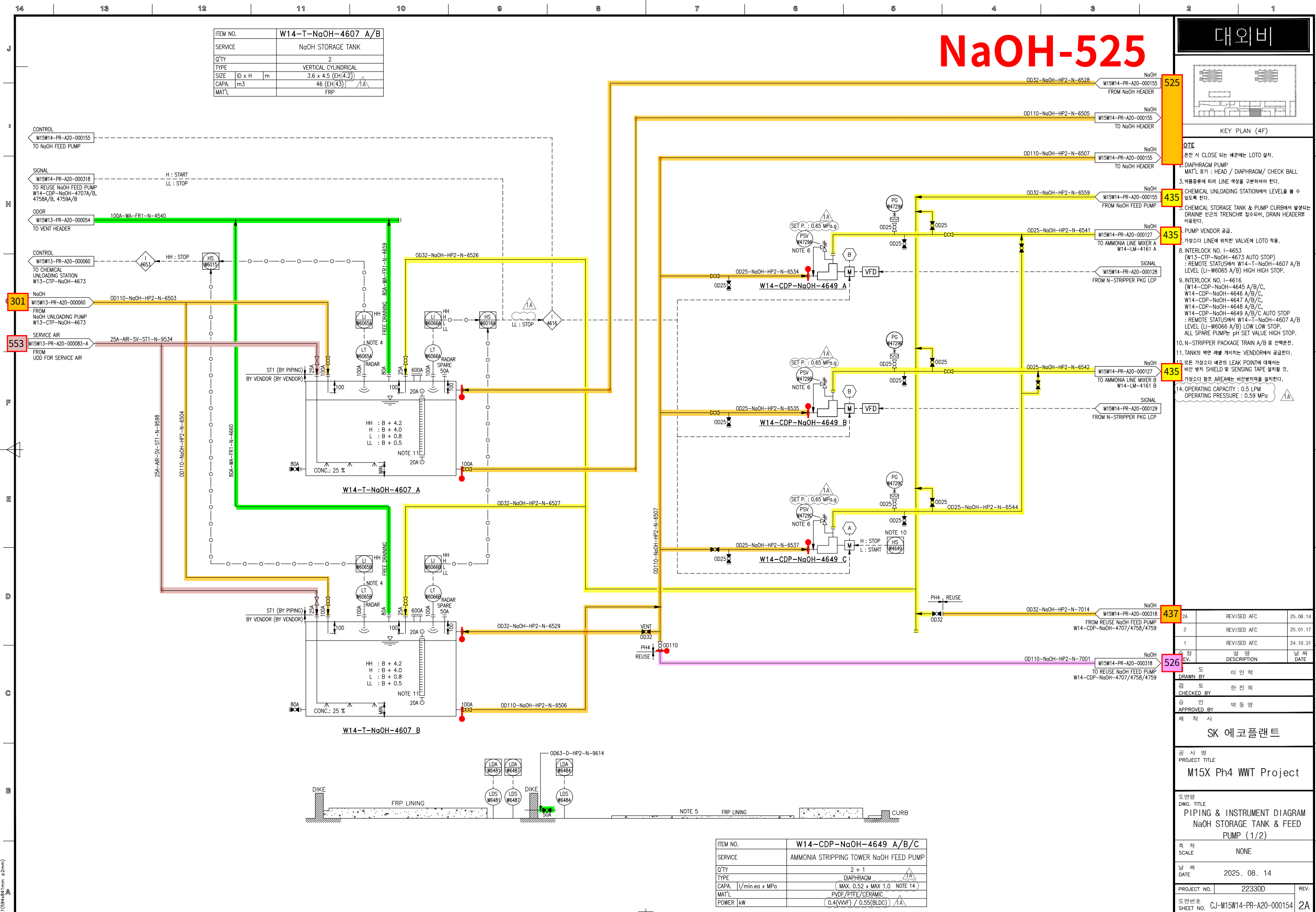
DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

NaOH-525

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	525	4FA	OD25-NaOH-HP2-N-6534-01	1 OF 1	0	CJ-M15W14-PR-A20-000154	2A	
2	525	4FA	OD25-NaOH-HP2-N-6535-01	1 OF 1	0	CJ-M15W14-PR-A20-000154	2A	
3	525	4FA	OD25-NaOH-HP2-N-6537-01	1 OF 1	0	CJ-M15W14-PR-A20-000154	2A	
4	525	4FA	OD32-NaOH-HP2-N-6508-01	1 OF 1	0	CJ-M15W14-PR-A20-000155	1A	
5	525	4FA	OD32-NaOH-HP2-N-6509-01	1 OF 1	0	CJ-M15W14-PR-A20-000155	1A	
6	525	4FA	OD32-NaOH-HP2-N-6510-01	1 OF 1	0	CJ-M15W14-PR-A20-000155	1A	
7	525	4FA	OD32-NaOH-HP2-N-6511-01	1 OF 1	0	CJ-M15W14-PR-A20-000155	1A	
8	525	4FA	OD32-NaOH-HP2-N-6512-01	1 OF 1	0	CJ-M15W14-PR-A20-000155	1A	
9	525	4FA	OD32-NaOH-HP2-N-6513-01	1 OF 1	0	CJ-M15W14-PR-A20-000155	1A	
10	525	4FA	OD32-NaOH-HP2-N-6514-01	1 OF 1	0	CJ-M15W14-PR-A20-000155	1A	
11	525	4FA	OD32-NaOH-HP2-N-6515-01	1 OF 1	0	CJ-M15W14-PR-A20-000155	1A	
12	525	4FA	OD32-NaOH-HP2-N-6516-01	1 OF 1	0	CJ-M15W14-PR-A20-000155	1A	
13	525	4FA	OD32-NaOH-HP2-N-6528-01	1 OF 1	0	CJ-M15W14-PR-A20-000154, 155	2A/1A	
14	525	4FA	OD32-NaOH-HP2-N-6529-01	1 OF 1	0	CJ-M15W14-PR-A20-000154	2A/1A	
15	525	4FA	OD32-NaOH-HP2-N-6531-01	1 OF 1	0	CJ-M15W14-PR-A20-000155	1A	
16	525	4FA	OD32-NaOH-HP2-N-6532-01	1 OF 1	0	CJ-M15W14-PR-A20-000155	1A	
17	525	4FA	OD32-NaOH-HP2-N-6533-01	1 OF 1	0	CJ-M15W14-PR-A20-000155	1A	
18	525	4FA	OD110-NaOH-HP2-N-6505-01	1 OF 1	0	CJ-M15W14-PR-A20-000154, 155	2A/1A	
19	525	4FA	OD110-NaOH-HP2-N-6506-01	1 OF 1	0	CJ-M15W14-PR-A20-000154	2A	
20	525	4FA	OD110-NaOH-HP2-N-6507-01	1 OF 1	0	CJ-M15W14-PR-A20-000154, 155	2A/1A	



대외비

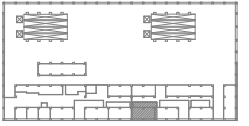
KEY PLAN (4F)

OTE
혼전 시 CLOSE 되는 배관에는 LOTO 설치.
2. DIAPHRAGM PUMP
MAT'L 표기 : HEAD / DIAPHRAGM / CHECK BALL
3. 약품종류에 따라 LINE 색상을 구분하여야 한다.
4. CHEMICAL UNLOADING STATION에서 LEVEL을 볼 수 있도록 한다.
5. CHEMICAL STORAGE TANK & PUMP CURB에서 발생하는 DRAINAGE 인단의 TRENCH로 흡수되어, DRAIN HEADER로 이용된다.
6. PUMP VENDOR 공급.
7. 가성소다 LINE에 위치한 VALVE에 LOTO 적용.
8. INTERLOCK NO. I-4653
(W13-CTP-NaOH-4673 AUTO STOP)
: REMOTE STATUS에서 W14-T-NaOH-4607 A/B LEVEL (U-W6065 A/B) HIGH HIGH STOP.
9. INTERLOCK NO. I-4616
(W14-CDP-NaOH-4645 A/B/C,
W14-CDP-NaOH-4646 A/B/C,
W14-CDP-NaOH-4647 A/B/C,
W14-CDP-NaOH-4648 A/B/C,
W14-CDP-NaOH-4649 A/B/C AUTO STOP
: REMOTE STATUS에서 W14-T-NaOH-4607 A/B LEVEL (LI-W6066 A/B) LOW LOW STOP.
ALL SPARE PUMPS는 pH SET VALUE HIGH STOP.
10. N-STRIPPER PACKAGE TRAIN A/B 중 선택연전.
11. TANK의 역관 레벨 경계치는 VENDOR에서 공급한다.
12. 모든 가성소다 배관의 LEAK 포인트에 대해서는 비산 방지 SHIELD 및 SENSING TAPE 설치할 것.
가성소다 환부 AREA에는 비산방지막을 설치한다.
14. OPERATING CAPACITY : 0.5 LPM
OPERATING PRESSURE : 0.59 MPa

2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31
수정 EV.	실명 DESCRIPTION	날짜 DATE
DRAWN BY 이인학		
CHECKED BY 한진희		
APPROVED BY 박동영		
제작사		
SK 에코플랜트		
공사명 PROJECT TITLE		
M15X Ph4 WWT Project		
도면명 DWG. TITLE		
PIPING & INSTRUMENT DIAGRAM NaOH STORAGE TANK & FEED PUMP (1/2)		
속척 SCALE		
NONE		
날짜 DATE		
2025. 08. 14		
PROJECT NO. 22330D		REV. A
도면번호 SHEET NO. CJ-M15W14-PR-A20-000154		2A

NaOH-525

대외비



KEY PLAN (4F)

NOTE

1. 운전 시 CLOSE 되는 배관에는 LOTO 실시.
2. DIAPHRAGM PUMP
MAT'L 표기 : HEAD / DIAPHRAGM/ CHECK BALL
3. 역류종류에 따라 LINE 역상을 구분하여야 한다.
4. CHEMICAL STORAGE TANK & PUMP CURB에서 발생하는 DRAIN은 인근의 TRENCH로 흡수되어, DRAIN HEADER로 이송된다.
5. PUMP VENDOR 공급.
6. NaOH LINE에 위치한 VALVE에 LOTO 적용.
7. HS-W1001A, W1001B 중 선택은전.
8. AIC-W2254, W2264 중 선택은전.
9. HS-W1902, W1903 중 선택은전.
10. HS-W1202, W1203 중 선택은전.
11. 모든 가성소터 배관의 LEAK POINT에는 비산 방지용 SHIELD 및 SENSING TAPE 설치할 것.
12. 가성소터 펌프 AREA에는 비산방지막을 설치한다.
13. OPERATING CAPACITY : 10 LPM
OPERATING PRESSURE : 0.59 MPa
14. OPERATING CAPACITY : 8 LPM
OPERATING PRESSURE : 0.59 MPa
15. OPERATING CAPACITY : 7 LPM
OPERATING PRESSURE : 0.59 MPa

ITEM NO.	W14-CDP-NaOH-4645 A/B/C
SERVICE	W.W NaOH FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA.	1/min.ea x MPa (MAX. 11.2 x MAX 0.7 NOTE 13)
MAT'L	PVDF/PTFE/CERAMIC
POWER kW	(0.75)

ITEM NO.	W14-CDP-NaOH-4646 A/B/C
SERVICE	CIF 2'nd NaOH FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA.	1/min.ea x MPa (MAX. 11.2 x MAX 0.7 NOTE 14)
MAT'L	PVDF/PTFE/CERAMIC
POWER kW	(0.75)

ITEM NO.	W14-CDP-NaOH-4647 A/B/C
SERVICE	HF 2'nd NaOH FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA.	1/min.ea x MPa (MAX. 11.2 x MAX 0.7 NOTE 15)
MAT'L	PVDF/PTFE/CERAMIC
POWER kW	(0.75)

ITEM NO.	W14-CDP-NaOH-4648 A/B/C
SERVICE	CIF ORGANIC NaOH FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA.	1/min.ea x MPa (MAX. 11.2 x MAX 0.7 NOTE 13)
MAT'L	PVDF/PTFE/CERAMIC
POWER kW	(0.75)

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정	실명	날짜
REV.	DESCRIPTION	DATE

제	도	이인학
DRAWN BY		
검	토	한진희
CHECKED BY		
인	박동영	
APPROVED BY		
제	작	사

SK 에코플랜트

공 사 명
PROJECT TITLE
M15X Ph4 WWT Project

도면명
DWG. TITLE
PIPING & INSTRUMENT DIAGRAM
NaOH STORAGE TANK & FEED PUMP
(2/2)

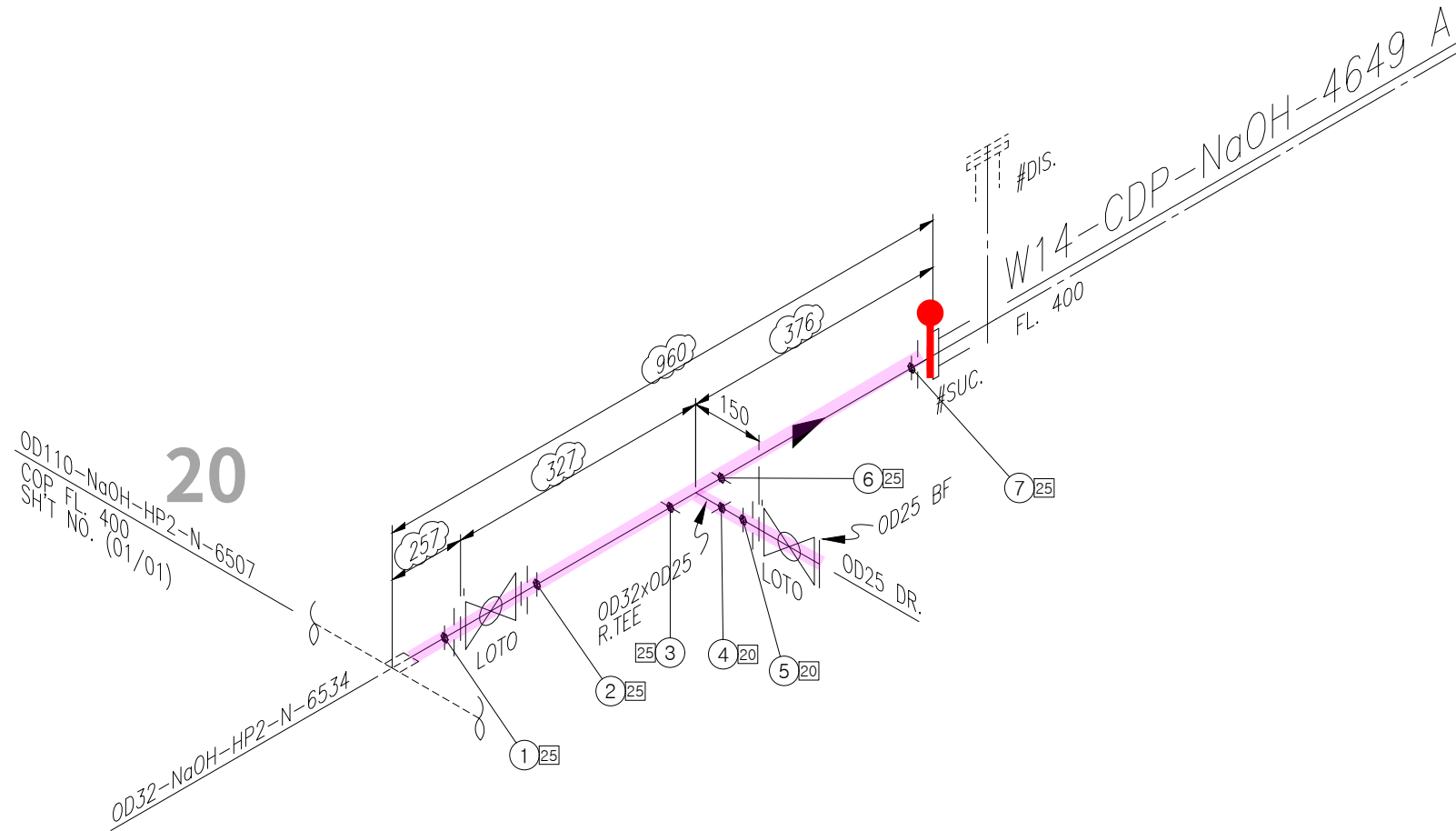
속 척
SCALE
NONE

날 짜
DATE
2025. 08. 14

PROJECT NO. 22330D
도면번호
SHEET NO. CJ-M15W14-PR-A20-000155 R1 P&ID FOR NaOH



ITEM	SIZE	Q'TY	DESCRIPTION
HP2			
P	OD25	0.9M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
T	OD25	1EA	TEE , HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
F	OD25	4 EA	FLG LAP JOINT / STUB END SET, KS 10K, FF,
FB	OD25	1EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-STSS304
			/ LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD25	4 EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD25	1EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B1	OD25	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VBLO	OD25	2 EA	BALL VALVE, KS 10K, FF FLG'D, BODY-PP /
			TRIM-PP / SEAT-PTFE, FB, LEVER + LOTO
SH	OD25	5EA	SHIELD



<NOTE>

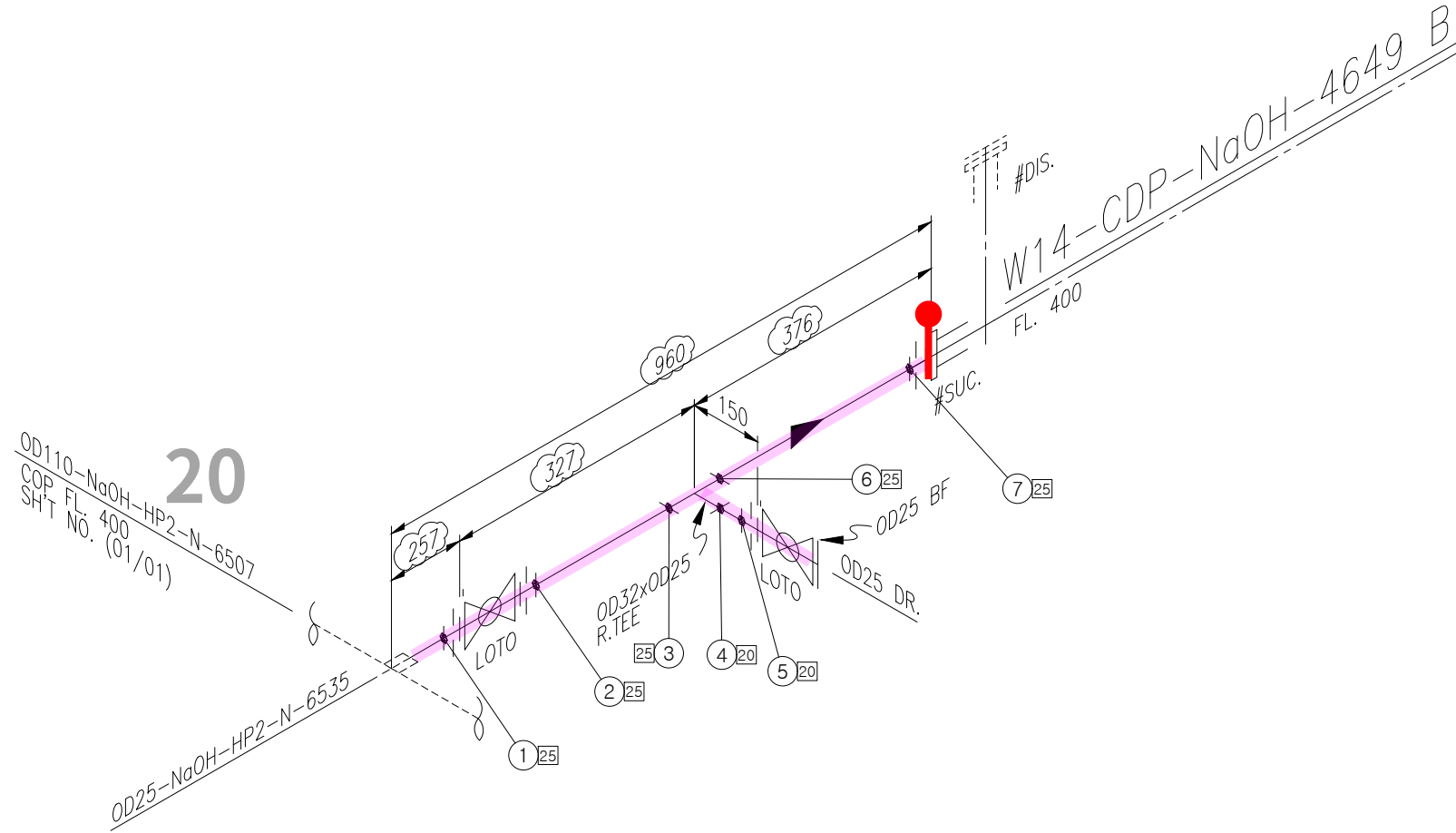
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	-	DESIGN	1	60	UNIT	4FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-040012-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	NaOH	22330D	
						REF. P & ID		REV.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000154		NO.						0D25-NaOH-HP2-N-6534	01 / 01	0



ITEM	SIZE	Q'TY	DESCRIPTION
		HP2	
P	OD25	1.0M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
T	OD25	1EA	TEE , HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
F	OD25	4 EA	FLG LAP JOINT / STUB END SET, KS 10K, FF,
FB	OD25	1EA	EAB BLIND-STEEL SET KS END; HDPE BLIND SETS304 / LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD25	4 EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD25	1EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B1	OD25	5SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VBLO	OD25	2 EA	BALL VALVE, KS 10K, FF FLC'D, BODY-PP /
			TRIM-PP / SEAT-PTFE, FB, LEVER + LOTO
SH	OD25	5EA	SHIELD



<NOTE>

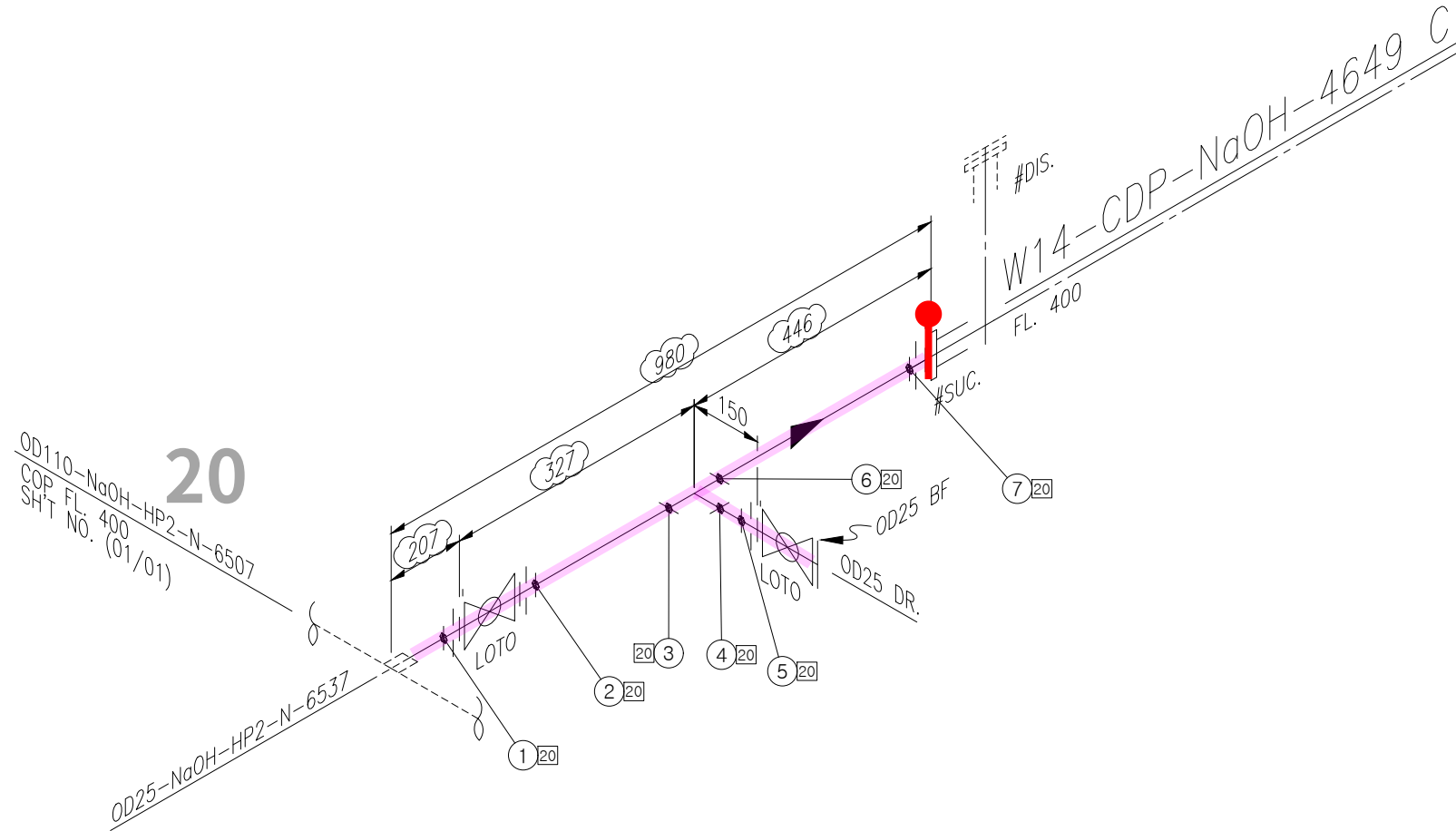
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1	60	UNIT	4FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-040012-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남 후	박동영	NaOH	22330D	
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000154								0D25-NaOH-HP2-N-6535	01 / 01	0



ITEM	SIZE	Q'TY	DESCRIPTION
		HP2	
P	OD25	1.0M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
T	OD25	1EA	TEE , HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
F	OD25	4 EA	FLG LAP JOINT / STUB END SET, KS 10K, FF,
FB	OD25	1EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-ST3304
			/ LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD25	4 EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD25	1EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B1	OD25	5SET	MACHINE BOLT & NYLON NUTS :ST 304 / W WASHERS
VBLO	OD25	2 EA	BALL VALVE, KS 10K, FF FLG'D, BODY-PP /
			TRIM-PP / SEAT-PTFE, FB, LEVER + LOTO
SH	OD25	5EA	SHIELD



<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB10/VB2/VB210	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD10	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						CONTRACTOR	PROJECT NAME	
	THICK.	-	DESIGN	1	60	UNIT	4FA							M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040012-A						COMMODITY		NaOH	JOB NO.
						REF. P & ID CJ-M15W14-PR-A20-000154	0	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	0D25-NaOH-HP2-N-6537	SH'T NO. 01 / 01
						REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D				

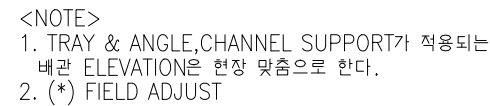


CLIENT



4

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	1	60	UNIT	4FA								M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040012-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남 후	박동영	COMMODITY	NaOH	JOB NO.	
						REF. P & ID	CJ-M15W14-PR-A20-000155							REV. NO.	DATE	DESCRIPTION	BY

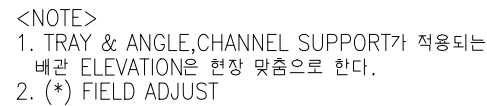


CLIENT



5

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1	60	UNIT	4FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-040012-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남 후	박동영	NaOH	22330D	
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000155								0D32-NaOH-HP2-N-6509	01 / 01	0

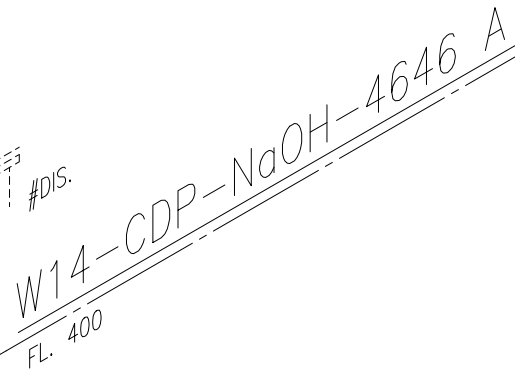


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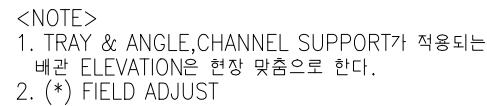
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INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	1	60	UNIT	4FA								M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040012-A							COMMODITY	NaOH	JOB NO.	
						REF. P & ID	CJ-M15W14-PR-A20-000155		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남 후	박동영	LINE NO.	OD32-NaOH-HP2-N-6510	SH'T NO.	REV.
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			01 / 01	0



* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

CLIENT  7



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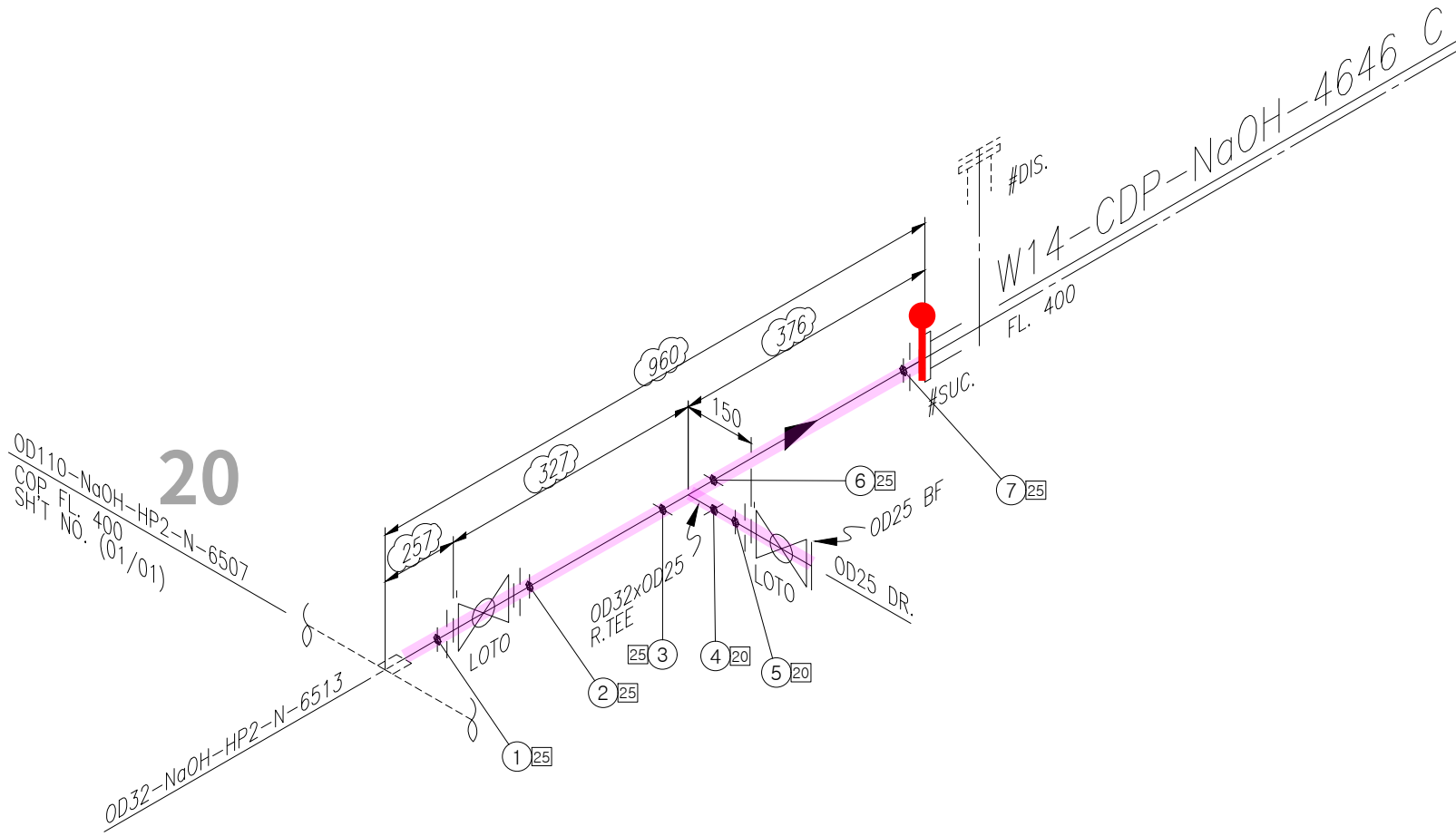


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INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1	60	UNIT	4FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-040012-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	NaOH	22330D	
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000155								0D32-NaOH-HP2-N-6512	01 / 01	0



ITEM	SIZE	Q'TY	DESCRIPTION
		HP2	
P	OD32	1.0M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
P	OD25	0.2M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
TR	OD32xOD25	1EA	TEE RED, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
F	OD32	3 EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STS304 / STUB END-HDPE PE100
F	OD25	1EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STS304 / STUB END-HDPE PE100
FB	OD25	1EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-STS304 / LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD32	3EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD25	1EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD25	1EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B1	OD32	3SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
B1	OD25	2SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VBLO	OD32	1EA	BALL VALVE, KS 10K, FF FLG'D, BODY-PP / TRIM-PP / SEAT-PTFE, FB, LEVER + LOTO
VBLO	OD25	1EA	BALL VALVE, KS 10K, FF FLG'D, BODY-PP / TRIM-PP / SEAT-PTFE, FB, LEVER + LOTO
SH	OD32	3EA	SHIELD
SH	OD25	2EA	SHIELD



<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT				9	
CONTRACTOR				PROJECT NAME M15X Ph4 Project	
COMMODITY NaOH				JOB NO. 22330D	
LINE NO. 0D32-NaOH-HP2-N-6513				SH'T NO. 01 / 01	REV. 0



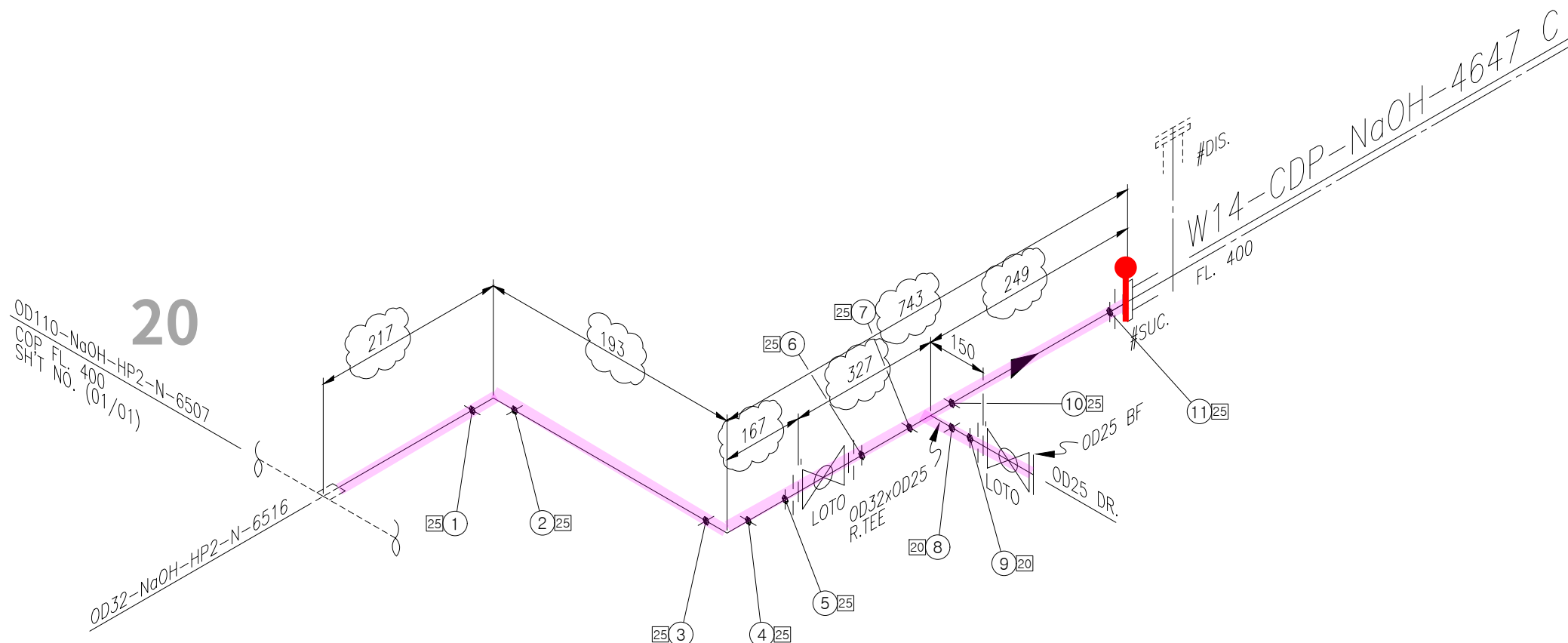
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CONTRACTOR 		PROJECT NAME M15X Ph4 Project	
COMMODITY NaOH		JOB NO. 22330D	
LINE NO. OD32-NaOH-HP2-N-6514		SH'T NO. 01 / 01	REV. 0



11

CONTRACTOR 		PROJECT NAME M15X Ph4 Project	
COMMODITY NaOH		JOB NO. 22330D	
LINE NO. 0D32-NaOH-HP2-N-6515	SH'T NO. 01 / 01		REV. 0

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<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1O/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VCS/VCS1/VCS2/VCS3/VCS4/VCS5/VCS6/VCS7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14								CONTRACTOR	PROJECT NAME
	THICK.	—	DESIGN	1	60	UNIT	4FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040012-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남 후	박동영	COMMODITY	NaOH	JOB NO.
						REF. P & ID	CJ-M15W14-PR-A20-000155							LINE NO.	0D32-NaOH-HP2-N-6516	SH'T NO.
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D		01 / 01	0

[illegible]



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14

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA W14							CONTRACTOR 	PROJECT NAME	
	THICK.	-	DESIGN	6.8	60	UNIT 4FA								M15X Ph4 Project	
			OPERATING	6.1	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040009-A							COMMODITY NaOH	JOB NO.	
								25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	22330D	
						REF. P & ID CJ-M15W14-PR-A20-000154	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	0D32-NaOH-HP2-N-6529	SH'T NO. 01 / 01	REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출될 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



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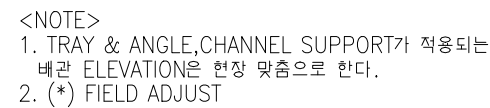
TOR  SK ecoplant

경고 : 피고 도역 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 무단으로 다른 사람에게 제공하여서는 안되며, 회사 내의 정보자의 폐해가 발생할 경우 정보자에게 민/형사상 책임을 요구할 수 있다.



16

CONTRACTOR 		PROJECT NAME M15X Ph4 Project	
COMMODITY NaOH		JOB NO. 22330D	
LINE NO. OD32-NaOH-HP2-N-6532		SH'T NO. 01 / 01	REV. 0



CLIENT



17

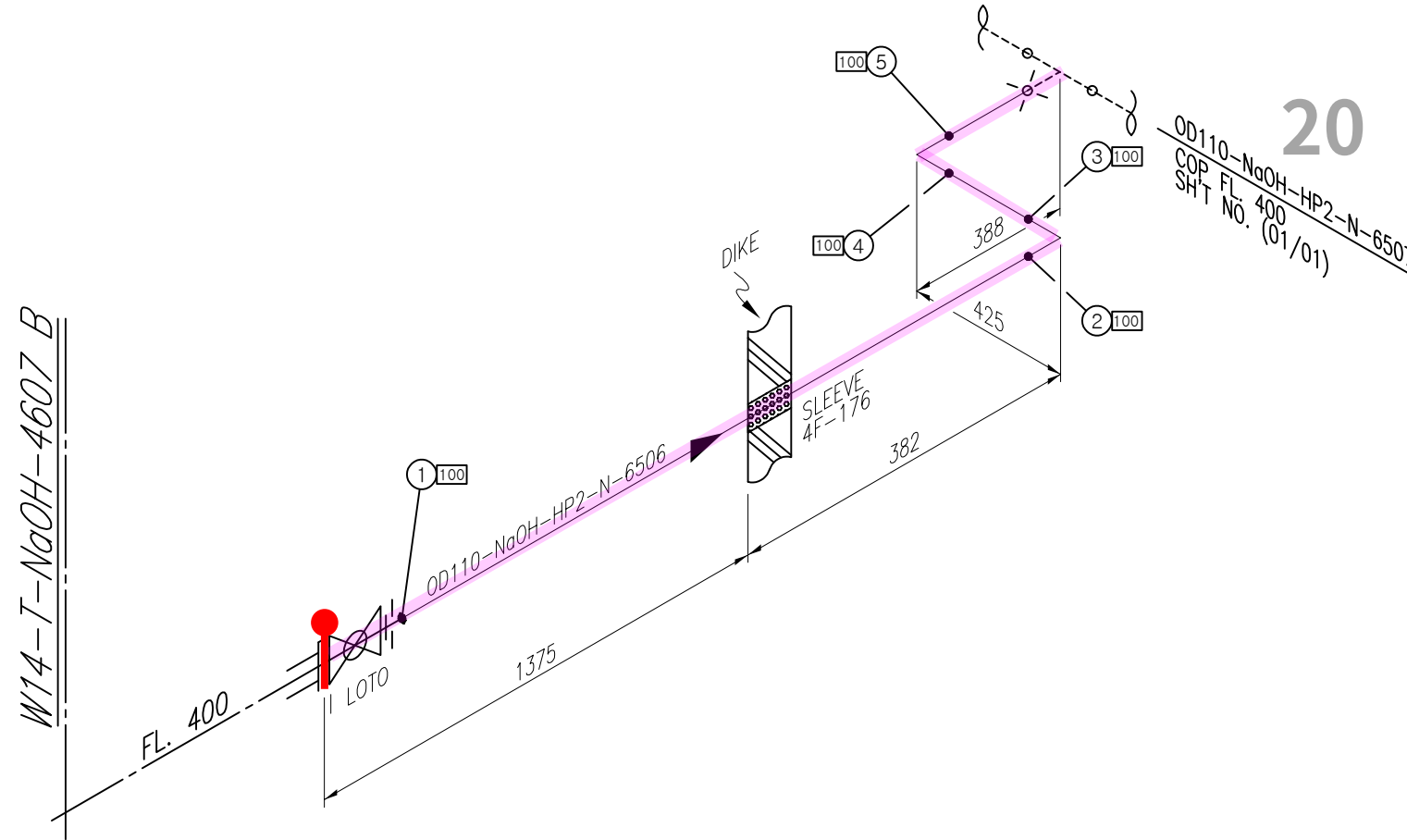
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1	60	UNIT	4FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-040012-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	NaOH	22330D	
						REF. P & ID		REV.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000155								0D32-NaOH-HP2-N-6533	01 / 01	0



CONTRACTOR 		PROJECT NAME M15X Ph4 Project	
COMMODITY NaOH		JOB NO. 22330D	
LINE NO. 0D110-NaOH-HP2-N-6505		SH'T NO. 01 / 01	REV. 0



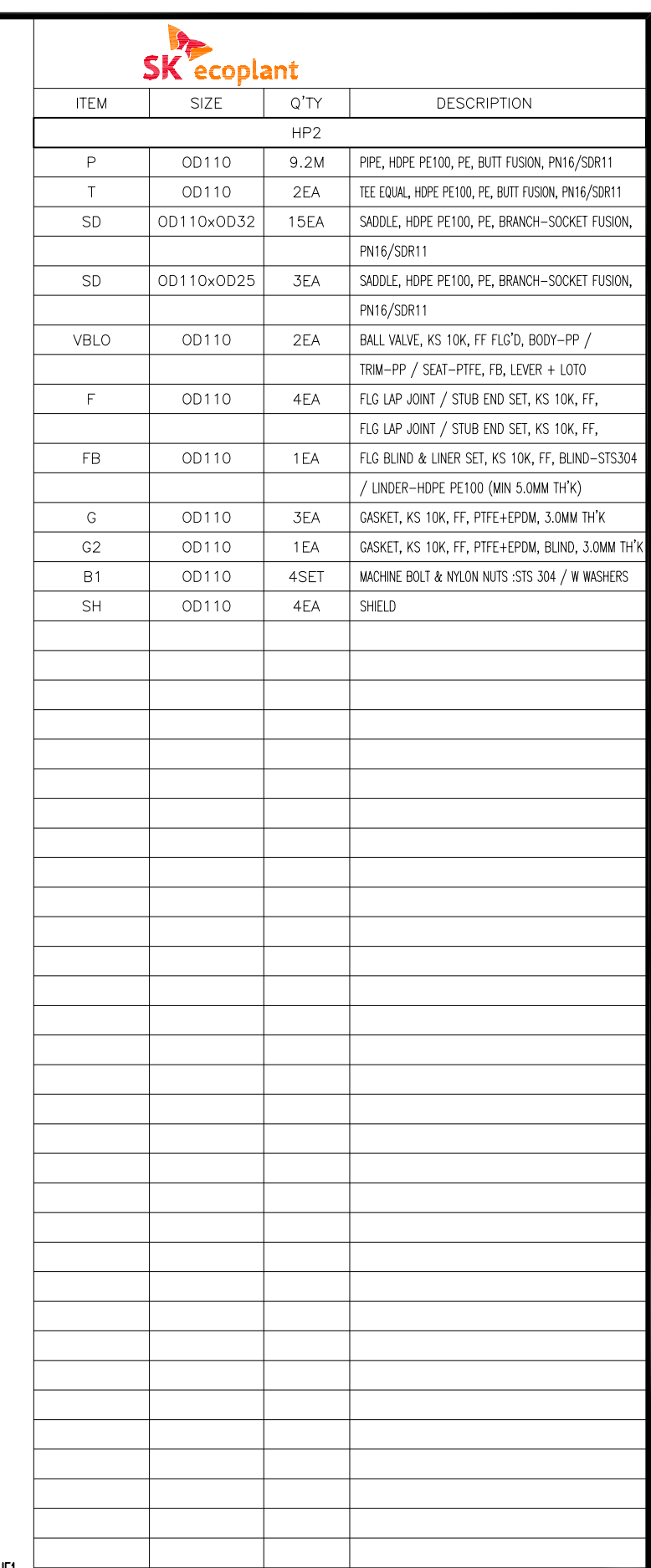
ITEM	SIZE	Q'TY	DESCRIPTION
		HP2	
P	OD110	2.6M	PIPE, HDPE PE100, PE, BUTT FUSION, PN16/SDR11
F	OD110	1EA	FLG LAP JOINT / STUB END SET, KS 10K, FF,
			LAP JOINT-STS304 / STUB END-HDPE PE100
G	OD110	2EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B1	OD110	2SET	MACHINE BOLT & NYLON NUTS :STS 304 / W WASHERS
VBLO	OD110	1EA	BALL VALVE, KS 10K, FF FLG'D, BODY-PP /
			TRIM-PP / SEAT-PTFE, FB, LEVER + LOTO
SH	OD110	2EA	SHIELD
E90	OD110	2EA	ELBOW 90 DEG, HDPE PE100, PE, BUTT FUSION, PN16/SDR11



<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA : VG/VG1/VG2	* BALL VA : VB/VB0/VB1/VB10/VB2/VB20	* BUTTERFLY VA : VF/VF4	* CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA : VL/VL1/VL2	* DIAPHRAGM VA : VD/VD0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHIELD : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14									CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1	60	UNIT	4FA										M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.									COMMODITY	NaOH	JOB NO.	
						CJ-M15W14-PI-A30-040012-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영			22330D		
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D		LINE NO.	OD110-NaOH-HP2-N-6506	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000154											01 / 01	0



CLIENT		20
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경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음